

Topology & Smooth Manifolds

Study Notes

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Updated regularly as the notes progress.

Introduction

These notes are a personal record of proofs and results encountered during my independent study of point-set topology and smooth manifold theory, following primarily the texts of Lee [1, 2] and Munkres [3]. They are not intended as a comprehensive treatment of either subject, but rather as a curated collection of arguments that I found instructive, surprising, or worth understanding in depth.

The selection reflects my background at the intersection of mathematics and theoretical physics. Many of the results here are foundational machinery for the analysis of differential operators on manifolds such as bump functions, exhaustion functions, partitions of unity which underlie constructions such as the heat kernel and the spectral theory of the Laplacian that appear in the physics notes on this site. Others, such as the results on topological groups, are included because they illuminate the structure of Lie groups, which are central to both modern geometry and theoretical physics.

Each section is self-contained and states all definitions needed for the proofs that follow. Cross-references are given where one result depends on another. Remarks at the end of each proof discuss the necessity of hypotheses, alternative approaches, and connections to broader theory.

A note on updates. These notes are a living document, updated regularly roughly weekly as my studies progress. Results are added section by section; later sections may appear in outline form before being completed. The current version covers:

- Smooth functions on manifolds and infinite dimensionality of $C^\infty(M, \mathbb{R})$
- Retracts of Hausdorff spaces
- Topological groups: open subgroups, generated subgroups, connected components

Forthcoming sections will treat the generalized gluing lemma, smooth exhaustion functions, partitions of unity, and introductory material on the heat kernel.

Feedback and corrections are welcome at ezzine@kth.se.

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I Smooth Functions and Infinite Dimensionality

We begin by recalling the basic objects. Throughout, M denotes a smooth manifold of dimension $n \geq 1$.

Definition 1.1 (Smooth Manifold). A *smooth manifold* of dimension n is a second-countable Hausdorff topological space M equipped with a maximal smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$, where each chart map $\varphi_\alpha: U_\alpha \rightarrow \mathbb{R}^n$ is a homeomorphism onto an open subset of $\mathbb{R}^n$ and all transition maps are smooth.

Definition 1.2 (Bump Function). Let M be a smooth manifold, $A \subseteq M$ a closed subset, and $U \supseteq A$ an open subset. A function $f \in C^\infty(M)$ is called a *bump function for A supported in U* if

$$0 \leq f \leq 1, \quad f|_A \equiv 1, \quad \text{supp}(f) \subseteq U.$$

Lemma 1.1 (Existence of Bump Functions). Let M be a smooth manifold. For any closed subset $A \subseteq M$ and any open set $U \supseteq A$, there exists a smooth bump function for A supported in U .

Proof. Consider the open cover of M given by $\mathcal{U} = \{U, M \setminus A\}$. Let $\{\psi_0, \psi_1\}$ be a smooth partition of unity subordinate to $\mathcal{U}$, so that $\text{supp}(\psi_0) \subseteq U$ and $\text{supp}(\psi_1) \subseteq M \setminus A$.

We claim ψ_0 is the desired bump function. By definition $0 \leq \psi_0 \leq 1$. For any $x \in A$ we have $x \notin M \setminus A$, so $x \notin \text{supp}(\psi_1)$ and thus $\psi_1(x) = 0$. The partition of unity condition $\psi_0(x) + \psi_1(x) = 1$ then gives $\psi_0(x) = 1$, so $\psi_0|_A \equiv 1$. Finally $\text{supp}(\psi_0) \subseteq U$ by construction. ■

Theorem 1.1. Let M be a smooth manifold with $\dim M \geq 1$. Then $C^\infty(M, \mathbb{R})$ is infinite dimensional as a real vector space.

Proof. Since M is a smooth manifold it contains a coordinate ball $B \subseteq M$ with homeomorphism $\varphi: B \xrightarrow{\sim} \varphi(B) \subseteq \mathbb{R}^n$. After translation and scaling we may assume $\varphi(B) = B_0(1)$, the open unit ball in $\mathbb{R}^n$.

Step 1: Disjoint open sets. For each $k \in \mathbb{N}^*$ define the open annular region

$$\mathcal{A}_k := B_0\left(\frac{1}{k}\right) \setminus \overline{B_0}\left(\frac{1}{k+1}\right) \subset \mathbb{R}^n,$$

and choose a point $x_k \in \mathcal{A}_k$ together with an open set $U_k \subset \mathcal{A}_k$ containing x_k . Since the annuli $\mathcal{A}_k$ are pairwise disjoint, the sets $\{U_k\}_{k \geq 1}$ are pairwise disjoint.

Step 2: Lifting to M . Set $W_k := \varphi^{-1}(U_k)$ and $p_k := \varphi^{-1}(x_k)$. Each W_k is open in M and the family $\{W_k\}$ is pairwise disjoint.

Step 3: Bump functions. For each k , choose a closed set C_k with $p_k \in C_k \subseteq W_k$. By Lemma 1.1 there exists $f_k \in C^\infty(M, \mathbb{R})$ with $f_k(p_k) = 1$ and $\text{supp}(f_k) \subseteq W_k$.

Step 4: Linear independence. Suppose $c_k \in \mathbb{R}$ satisfy $\sum_{k=1}^{\infty} c_k f_k(x) = 0$ for all $x \in M$. For fixed $j \in \mathbb{N}^*$, evaluating at p_j : since $\text{supp}(f_k) \subseteq W_k$ and $W_j \cap W_k = \emptyset$ for $k \neq j$, we have $f_k(p_j) = 0$ for $k \neq j$. Therefore

$$0 = \sum_{k=1}^{\infty} c_k f_k(p_j) = c_j f_j(p_j) = c_j.$$

Since j was arbitrary all coefficients vanish, so $\{f_k\}_{k \geq 1}$ is linearly independent and $C^\infty(M, \mathbb{R})$ is infinite dimensional. ■

Remark. The annular construction gives a completely explicit linearly independent family, which is more concrete than the standard argument that merely invokes the existence of infinitely many points. The Hausdorff property of M is used implicitly in Step 3 to find the closed sets $C_k \subset W_k$.

2 Retracts of Hausdorff Spaces

Definition 2.1 (Retract). Let X be a topological space and $A \subseteq X$. We say A is a *retract* of X if there exists a continuous map $r: X \rightarrow A$ with $r|_A = \text{id}_A$. The map r is called a *retraction*.

Theorem 2.1. Let X be a Hausdorff topological space and $A \subseteq X$ a retract of X . Then A is closed in X .

Proof. Let $r: X \rightarrow A$ be a retraction. We show $X \setminus A$ is open by exhibiting, for each $p \in X \setminus A$, an open neighbourhood of p disjoint from A .

Step 1: Separation. Since $p \notin A$ and $r(p) \in A$ we have $p \neq r(p)$. As X is Hausdorff there exist disjoint open sets $U, W \subseteq X$ with $p \in U$ and $r(p) \in W$.

Step 2: A neighbourhood of p disjoint from A . Since $r(p) \in W \cap A$, the set $W \cap A$ is nonempty. By continuity of r the preimage $r^{-1}(W \cap A)$ is open. Set $V := U \cap r^{-1}(W \cap A)$. Then V is open and $p \in V$: indeed $p \in U$ and $r(p) \in W \cap A$ gives $p \in r^{-1}(W \cap A)$.

Step 3: $V \cap A = \emptyset$. Suppose $a \in V \cap A$. Then $a \in r^{-1}(W \cap A)$, so $r(a) \in W$. But $a \in A$ and r is a retraction, so $r(a) = a$, giving $a \in W$. Also $a \in V \subseteq U$, so $a \in U \cap W = \emptyset$, a contradiction. Hence $V \subseteq X \setminus A$, and A is closed. ■

Remark. An equivalent proof uses the equaliser characterisation: $A = \{x \in X : r(x) = x\}$ is the preimage of the diagonal $\Delta_X \subset X \times X$ under the continuous map $x \mapsto (r(x), x)$. Since X is Hausdorff, Δ_X is closed, hence so is A . Both arguments use Hausdorffness in an essential way; the result fails in non-Hausdorff spaces.

3 Topological Groups

Definition 3.1 (Topological Group). A *topological group* is a group G equipped with a topology such that the multiplication map $(g, h) \mapsto gh$ and the inversion map $g \mapsto g^{-1}$ are both continuous.

We begin with a basic but repeatedly used observation.

Lemma 3.1 (Left Translation is a Homeomorphism). For any $g \in G$, the left multiplication map

$$L_g: G \rightarrow G, \quad p \mapsto gp,$$

is a homeomorphism. In particular, if $U \subseteq G$ is open then so is every left coset $gU := L_g(U)$.

Proof. L_g is continuous by the group axioms, and $(L_g)^{-1} = L_{g^{-1}}$ is likewise continuous. Hence L_g is a homeomorphism, and the image of any open set under a homeomorphism is open. ■

Proposition 3.1. Let G be a topological group. Every open subgroup $H \leq G$ is also closed.

Proof. By Lemma 3.1 every left coset gH is open. The left cosets of H partition G , so

$$G \setminus H = \bigcup_{g \in G \setminus H} gH$$

is a union of open sets, hence open. Therefore H is closed. ■

Remark. The partition of G into cosets of H is a partition into clopen sets whenever H is open. In particular, if G is connected the only open subgroup is G itself. Note the converse fails: $\mathbb{Z} \leq \mathbb{R}$ is closed but not open.

Proposition 3.2. Let G be a topological group and U an open neighbourhood of the identity $1 \in G$. Then $\langle U \rangle$ is open and closed in G .

Proof. Recall $\langle U \rangle = \{u_1^{\varepsilon_1} \cdots u_n^{\varepsilon_n} \mid n \in \mathbb{N}, u_i \in U, \varepsilon_i \in \{1, -1\}\}$.

$\langle U \rangle$ is open. We show $\langle U \rangle = \bigcup_{g \in \langle U \rangle} gU$.

($\subseteq$): For $p \in \langle U \rangle$, since $1 \in U$ we have $p = p \cdot 1 \in pU \subseteq \bigcup_{g \in \langle U \rangle} gU$.

($\supseteq$): If $k \in \ell U$ for some $\ell \in \langle U \rangle$ then $k = \ell u$ with $u \in U \subseteq \langle U \rangle$, so $k \in \langle U \rangle$.

Each coset gU is open by Lemma 3.1, so $\langle U \rangle$ is a union of open sets, hence open.

$\langle U \rangle$ is closed. It is an open subgroup, so Proposition 3.1 applies. ■

Remark. If U is merely a neighbourhood rather than an open set, replace it by an open set $V \subseteq U$ with $1 \in V$; then $\langle V \rangle \subseteq \langle U \rangle$ and the result applies to $\langle V \rangle$.

Lemma 3.2. Let $\{A_\alpha\}$ be a family of connected subsets of a topological space X with $\bigcap_\alpha A_\alpha \neq \emptyset$. Then $\bigcup_\alpha A_\alpha$ is connected.

Proof. Let $f: \bigcup_\alpha A_\alpha \rightarrow \{0, 1\}$ be locally constant and fix $x_0 \in \bigcap_\alpha A_\alpha$. Each $f|_{A_\alpha}$ is constant (as A_α is connected), equal to $f(x_0)$. Hence $f \equiv f(x_0)$ on the whole union, which is therefore connected. ■

Lemma 3.3. The product of two connected spaces is connected.

Proof. Standard; see [3], Theorem 23.6. ■

Proposition 3.3. Let G be a topological group and $U \subseteq G$ a connected subset with $1 \in U$. Then $\langle U \rangle$ is connected.

Proof. *Step 1:* U^n is connected for all $n \in \mathbb{N}^*$. Here $U^n = \{u_1 \cdots u_n \mid u_i \in U\}$. For $n = 1$ this is the hypothesis. If U^n is connected then $U^n \times U$ is connected by Lemma 3.3, and $U^{n+1} = \mu(U^n \times U)$ is the continuous image of a connected set, hence connected.

Step 2: $V := U \cup U^{-1}$ is connected. The inversion map is continuous, so $U^{-1} = \{u^{-1} \mid u \in U\}$ is connected. Since $1 \in U \cap U^{-1}$, Lemma 3.2 gives that $V = U \cup U^{-1}$ is connected.

Step 3: $\langle U \rangle$ is connected. By Step 1 applied to V , each V^n is connected and $1 \in V^n$ for all n . Any element of $\langle U \rangle$ is a product of elements of V , so $\langle U \rangle = \bigcup_{n \geq 1} V^n$. Since all V^n share the point 1, Lemma 3.2 gives that $\langle U \rangle$ is connected. ■

Proposition 3.4. Let G be a connected topological group. Then every connected open neighbourhood U of 1 satisfies $\langle U \rangle = G$.

Proof. By Proposition 3.2, $\langle U \rangle$ is open and closed. It is nonempty since $1 \in \langle U \rangle$. A nonempty clopen subset of a connected space must be the whole space, so $\langle U \rangle = G$. ■

Remark. In the setting of Lie groups, the exponential map $\exp: \mathfrak{g} \rightarrow G$ is a local diffeomorphism near $0 \in \mathfrak{g}$, so its image contains a connected open neighbourhood of $1 \in G$. Proposition 3.4 then implies that every connected Lie group is (algebraically) generated by the image of $\exp$, a foundational fact in Lie theory.

References

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- [3] J. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.
- [4] S. Rosenberg, *The Laplacian on a Riemannian Manifold*, Cambridge University Press, 1997.